# Ethical Hacking and Countermeasures
## Wireshark Filters Cheat Sheet

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**Wireshark**
Source: https://www.wireshark.org

Wireshark is a protocol analyzer tool that can capture and interactively browse the traffic running on a computer network. This tool uses Winpcap to capture packets on its supported networks. It captures live network traffic from Ethernet, IEEE 802.11, PPP/HDLC, ATM, Bluetooth, USB, Token Ring, Frame Relay, FDDI networks. The captured files can be programmatically edited via the command-line. A set of filters for customized data displays can be refined using a display filter.

### 1. Frame Filters

| Wireshark Filters | Description |
|---|---|
| frame.cap_len | Frame length stored into the capture file |
| -B, --source-port | Coloring Rule Name |
| frame.coloring_rule.string | Coloring Rule String |
| frame.file_off | File Offset |
| frame.number | Frame Number |
| frame.p2p_dir | Point-to-Point Direction |
| frame.protocols | Protocols in frame |
| frame.ref_time | Time Reference frame |
| frame.time | Arrival Time |
| frame.ignored | Frame is ignored |
| frame.len | Frame length on the wire |
| frame.link_nr | Link Number |
| frame.marked | Frame is marked |
| frame.md5_hash | Frame MD5 Hash |
| frame.time_delta | Time delta from previous captured frame |
| frame.time_delta_displayed | Time delta from previous displayed frame |
| frame.time_epoch | Epoch Time |
| frame.time_invalid | Arrival Time: Fractional second out of range (0-1000000000) |

### 2. Ethernet Filters

| Wireshark Filters | Description |
|---|---|
| eth.addr | Address of Ethernet or other MAC address |
| eth.dst | Destination of Ethernet or other MAC address |
| eth.ig | IG bit |
| eth.len | Length |
| eth.lg | LG bit |
| eth.src | Source of Ethernet or other MAC address |
| eth.trailer | Trailer |
| eth.type | Type |
| eth.src.oui_resolved | Source OUI (resolved) |
| eth.vlan.id | VLAN |
| eth.fcs_bad | Bad checksum |

### 3. DNS Filters

| Wireshark Filters | Description |
|---|---|
| dns.afsdb.hostname | Display DNS hostname |
| dns.a | Display IP address of DNS server |
| dns.caa.tag_length | Tag length |
| dns.csync.flags | Flags |
| dns.count.prerequisites | Prerequisites |

### 4. ARP Filters

| Wireshark Filters | Description |
|---|---|
| arp.duplicate-address-detected | Locate duplicate IP address traffic |
| arp.proto.size | Display protocol size |
| Display protocol size | Display protocol type |
| arp.src.hw_mac | Display sender's MAC address |
| arp.src.proto | Display sender's protocol address |
| arp.src.proto_ipv4 | Display sender's IP address |
| arp.dst.hw_mac | Target MAC address |
| arp.dst.proto_ipv4 | Target IP address |
| arp.hw.size | Hardware size |
| arp.hw.type | Hardware type |
| arp.opcode | Opcode |

### 5. PPP Filters

| Command | Description |
|---|---|
| ppp.address | Address |
| ppp.control | Control |
| ppp.direction | Direction |
| ppp.protocol | Protocol |

### 6. DTP Filters

| Command | Description |
|---|---|
| dtp.neighbor | Neighbor |
| dtp.tlv_type | Type |
| dtp.tlv_len | Length |
| dtp.version | Version |
| dtp.domain | Domain |
| dtp.senderid | Sender ID |
| dtp.tos | Trunk Operating Status |
| dtp.truncated | DTP message is truncated prematurely |
| dtp.data | Data |
| dtp.tas | Trunk Administrative Status |
| dtp.tat | Trunk Administrative Type |

### 7. IEEE 802.1Q VLAN Filters

| Command | Description |
|---|---|
| vlan.trailer | Trailer |
| vlan.priority | Priority |
| vlan.len | Length |
| vlan.cfi | CFI |
| vlan.etype | Type |
| vlan.id | ID |
| vlan.too_many_tags | Too many nested VLAN tags |
| vlan.len.past_end | Length field value goes past the end of the payload |



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### 8. MPLS Filters

| Wireshark Filters | Description |
|---|---|
| mpls.bottom | MPLS Bottom of Label Stack |
| mpls.cw.control | MPLS Control Channel |
| mpls.cw.res | Reserved |
| mpls.exp | MPLS Experimental Bits |
| mpls.label | MPLS Label |
| mpls.oam.bip16 | BIP16 |
| mpls.oam.defect_location | Defect Location (AS) |
| mpls.oam.defect_type | Defect Type |
| mpls.oam.frequency | Frequency |
| mpls.oam.function_type | Function Type |
| mpls.oam.ttsi | Trail Termination Source Identifier |
| mpls.ttl | MPLS TTL |

### 9. VTP (VLAN Trunking Protocol) Filters

| Wireshark Filters | Description |
|---|---|
| vtp.code | Code |
| vtp.conf_rev_num | Configuration Revision Number |
| vtp.followers | Followers |
| vtp.md | Management Domain |
| vtp.md5_digest | MD5 Digest |
| vtp.md_len | Management Domain Length |
| vtp.neighbor | Neighbor |
| vtp.seq_num | Sequence Number |
| vtp.start_value | Start Value |
| vtp.upd_id | Updater Identity |
| vtp.upd_ts | Update Timestamp |
| vtp.version | Version |
| vtp.vlan_info.802_10_index | 802.10 Index |
| vtp.vlan_info.isl_vlan_id | ISL VLAN ID |
| vtp.vlan_info.len | VLAN Information Length |
| vtp.vlan_info.mtu_size | MTU Size |
| vtp.vlan_info.status.vlan_susp | VLAN suspended |
| vtp.vlan_info.tlv_len | Length |
| vtp.vlan_info.tlv_type | Type |
| vtp.vlan_info.vlan_name | VLAN Name |
| vtp.vlan_info.vlan_name_len | VLAN Name Length |
| vtp.vlan_info.vlan_type | VLAN Type |

### 10. FR (Frame Relay) Filters

| Wireshark Filters | Description |
|---|---|
| fr.becn | BECN |
| fr.chdlctype | Type |
| fr.control | Control Field |
| fr.control.f | Final |
| fr.control.ftype | Frame type |
| fr.control.p | Poll |
| fr.control.s_ftype | Supervisory frame type |
| fr.control.u_modifier_cmd | Command |
| fr.control.u_modifier_resp | Response |
| fr.upper_dlci | Upper DLCI |
| fr.third_dlci | Third DLCI |
| fr.snaptype | Type |
| fr.snap.pid | Protocol ID |
| fr.snap.oui | Organization Code |
| fr.second_dlci | Second DLCI |
| fr.lower_dlci | Lower DLCI |
| fr.dlcore_control | DL-CORE Control |

### 11. IPv4 Filters

| Wireshark Filters | Description |
|---|---|
| ip.addr | Filter by Single IP Address displays all traffic from IP, be it source or destination |
| ip.dst | Filter by destination display traffic only from IP destination |
| ip.addr== <IP address> && tcp. port= <port number> | Monitoring the Specific Port using the IP address and TCP port number. |
| ip.addr == <IP address1> or ip.addr == <IP address2> | Filter by Multiple IP Addresses |
| ip.checksum.status | Display the header checksum status |
| ip.src != xxx.xxx.xxx.xxx && ip.dst != xxx.xxx.xxx.xxx && sip | Filter by a protocol (e.g. SIP) and filter out unwanted IPs |
| ip.src | Filter by Source IP display traffic only from IP source |
| ip.checksum | Header checksum |
| ip.checksum_bad | Bad |
| ip.checksum_good | Good |
| ip.dsfield | Differentiated Services Field |
| ip.dsfield.ect | ECN-Capable Transport (ECT) |
| ip.dst | Destination |
| ip.dst_host | Destination Host |
| ip.flags | Flags |
| ip.flags.df | Do not fragment |
| ip.flags.mf | More fragments |
| ip.flags.rb | Reserved bit |
| ip.frag_offset | Fragment offset |
| ip.fragment.error | Defragmentation error |
| ip.fragment.multipletails | Multiple tail fragments found |
| ip.fragment.overlap | Fragment overlap |
| ip.fragment.overlap.conflict | Conflicting data in fragment overlap |
| ip.fragment.toolongfragment | Fragment too long |
| ip.fragments | IPv4 Fragments |
| ip.hdr_len | Header Length |
| ip.host | Source or Destination Host |

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| | |
|---|---|
| ip.id | Identification |
| ip.len | Total Length |
| ip.proto | Protocol |
| ip.reassembled_in | Reassembled IPv4 in frame |
| ip.src | Filter by Source IPv4 |
| ip.src_host | Source Host |
| ip.tos | Type of Service |
| ip.tos.cost | Cost |
| ip.tos.delay | Delay |
| ip.tos.precedence | Precedence |
| ip.tos.reliability | Reliability |
| ip.tos.throughput | Throughput |
| ip.ttl | Time to live |
| ip.version | Version |

### 12. IPv6 Filters

| Wireshark Filters | Description |
|---|---|
| ipv6.addr | Source or Destination Address |
| ipv6.class | Traffic class |
| ipv6.dst | Destination |
| ipv6.dst_host | Destination Host |
| ipv6.dst_opt | Destination Option |
| ipv6.flow | Flow Label |
| ipv6.fragment | IPv6 Fragment |
| ipv6.fragment.error | Defragmentation error |
| ipv6.hop_opt | Hop-by-Hop Option |
| ipv6.host | Source or Destination Host |
| ipv6.mipv6_home_address | Home Address |
| ipv6.mipv6_length | Option Length |
| ipv6.mipv6_type | Option Type |
| ipv6.nxt | Next Header |
| ipv6.fragment.more | More Fragment |
| ipv6.fragment.multipletails | Multiple tail fragments found |
| ipv6.fragment.offset | Offset |
| ipv6.fragment.overlap | Fragment overlap |
| ipv6.fragment.overlap.conflict | Conflicting data in fragment overlap |
| ipv6.fragment.toolongfragment | Fragment too long |
| ipv6.fragments | IPv6 Fragments |
| ipv6.fragment.id | Identification |
| ipv6.hlim | Hop Limit |
| ipv6.plen | Payload Length |
| ipv6.reassembled_in | Reassembled IPv6 in frame |
| ipv6.routing_hdr | Routing Header, Type |
| ipv6.routing_hdr.addr | Address |
| ipv6.routing_hdr.left | Segments Left |
| ipv6.routing_hdr.type | Type |
| ipv6.src_host | Source Host |
| ipv6.version | Version |

### 13. ICMP Filters

| Wireshark Filters | Description |
|---|---|
| icmp.checksum_bad | Display bad checksum |
| icmp.type==8 or icmp.type==0 | Detect an ICMP ping sweep attempt |
| icmp.checksum | Display checksum |
| icmp.ident | Display identifier (BE) |
| icmp.seq | Display sequence number (BE) |
| icmp.mtu | Display MTU of next hop |
| icmp.redir_gw | Display Gateway address |
| icmp.checksum | Checksum |
| icmp.code | Code |
| icmp.type | Type |

### 14. ICMPv6 Filters

| Wireshark Filters | Description |
|---|---|
| icmpv6.all_comp | All components |
| icmpv6.checksum | Checksum |
| icmpv6.checksum_bad | Bad checksum |
| icmpv6.code | Code |
| icmpv6.comp | Component |
| icmpv6.haad.ha_addrs | Home agent addresses |
| icmpv6.identifier | Identifier |
| icmpv6.option | ICMPv6 Option |
| icmpv6.option.cga | CGA |
| icmpv6.option.length | Length |
| icmpv6.option.name_type | Name type |
| icmpv6.option.name_type.fqdn | FQDN |
| icmpv6.option.name_x501 | DER Encoded X.501 Name |
| icmpv6.option.rsa.key_hash | Key hash |
| icmpv6.option.type | Type |
| icmpv6.ra.cur_hop_limit | Cur hop limit |
| icmpv6.ra.reachable_time | Reachable time |
| icmpv6.ra.retrans_timer | Retrans timer |
| icmpv6.ra.router_lifetime | Router lifetime |
| icmpv6.recursive_dns_serv | Recursive DNS Servers |
| icmpv6.type | Type |

### 15. TCP Filters

| Wireshark Filters | Description |
|---|---|
| tcp contains traffic | Displays all TCP packets that contains the word 'traffic' |
| tcp.port==23 | Monitoring the Specific Port using tcp port number. |
| tcp.flags.reset==1 | Displays all TCP resets |
| tcp.analysis. retransmission | Displays all retransmissions in the trace |
| tcp.port == 4000 | Sets a filter for any TCP packet with 4000 as a source or destination port |

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| | |
|---|---|
| tcp.port eq 25 or icmp | Displays only SMTP (port 25) and ICMP traffic |
| tcp.dstport==7 | Detect TCP ping sweep attempt |
| tcp.flags==0X029 | View the packets with FIN, PSH, and URG TCP flags set |
| tcp.ack | Acknowledgment number |
| tcp.checksum | Checksum |
| tcp.checksum _ bad | Bad Checksum |
| tcp.checksum _ good | Good Checksum |
| tcp.options.qs | TCP QS Option |
| tcp.options.sack | TCP SACK Option |
| tcp.options.sack _ le | TCP SACK Left Edge |
| tcp.options.sack _ perm | TCP SACK Permitted Option |
| tcp.continuation _ to | This is a continuation to the PDU in frame |
| tcp.options.sack _ re | TCP SACK Right Edge |
| tcp.options.time _ stamp | TCP Time Stamp Option |
| tcp.flags | Flags |
| tcp.dstport | Destination Port |
| tcp.options.wscale | TCP Window Scale Option |
| tcp.flags.ack | Acknowledgment |
| tcp.options.wscale _ val | TCP Windows Scale Option Value |
| tcp.flags.cwr | Congestion Window Reduced (CWR) |
| tcp.pdu.last _ frame | Last frame of this PDU |
| tcp.flags.ecn | ECN-Echo |
| tcp.pdu.size | PDU Size |
| tcp.flags.fin | Fin |
| tcp.pdu.time | Time until the last segment of this PDU |
| tcp.flags.push | Push |
| tcp.port | Source or Destination Port |
| tcp.flags.reset | Reset |
| tcp.flags.syn | TCP; Filter SYN flag |
| tcp.flags.urg | Urgent |
| tcp.hdr _ len | Header Length |
| tcp.segment | TCP Segment |
| tcp.segment.error | Reassembling error |
| tcp.segment.multipletails | Multiple tail segments found |
| tcp.len | TCP Segment Len |
| tcp.nxtseq | Next sequence number |
| tcp.options | TCP Options |
| tcp.options.cc | TCP CC Option |
| tcp.options.ccecho | TCP CC Echo Option |
| tcp.options.ccnew | TCP CC New Option |
| tcp.options.echo | TCP Echo Option |
| tcp.options.echo _ reply | TCP Echo Reply Option |
| tcp.options.md5 | TCP MD5 Option |
| tcp.options.mss | TCP MSS Option |
| tcp.options.mss _ val | MSS Value |
| tcp.segment.overlap | Segment overlap |
| tcp.segment.overlap.conflict | Conflicting data in segment overlap |
| tcp.segment.toolongfragment | Segment too long |
| tcp.segments | Reassembled TCP Segments |
| tcp.seq | Sequence number |
| tcp.srcport | Source Port |
| tcp.time _ delta | Time since previous frame in this TCP stream |
| tcp.time _ relative | Time since first frame in this TCP stream |
| tcp.urgent _ pointer | Urgent pointer |
| tcp.window _ size | Calculated window size |

### 12. IPv6 Filters

| Wireshark Filters | Description |
|---|---|
| udp contains 33:27:5 | Sets a filter for the HEX values of 0x33 0x27 0x58 at any offset |
| udp.dstport==7 | Detect UDP ping sweep attempt |
| udp.proc.dstuid | Displays the destination process User ID |
| udp.proc.srcuid | Displays the source process User ID |
| udp.checksum | Checksum |
| udp.dstport | Destination Port |
| udp.srcport | Source Port |
| udp.checksum _ bad | Bad checksum |
| udp.length | Length |
| udp.checksum.zero | Illegal Checksum value (0) |
| udp.pdu.size | PDU Size |
| udp.possible _ traceroute | Possible traceroute |
| udp.proc.dstpid | Destination process ID |
| udp.stream | Stream index |
| udp.time _ delta | Time since previous frame |
| udp.time _ relative | Time since first frame |
| udplite.checksum _ coverage.bad | Bad checksum coverage length value |

### 17. HTTP Filters

| Wireshark Filters | Description |
|---|---|
| http.request | Displays and filter all HTTP GET requests |
| http.accept | Accept |
| http.accept _ encoding | Accept Encoding |
| http.accept _ language | Accept-Language |
| http.authbasic | Credentials |
| http.authcitrix | Citrix AG Auth |
| http.authcitrix.domain | Citrix AG Domain |
| http.authcitrix.password | Citrix AG Password |
| http.authcitrix.session | Citrix AG Session ID |
| http.authcitrix.user | Citrix AG Username |
| http.authorization | Authorization |
| http.referer | Referer |
| http.cache _ control | Cache-Control |
| icmpv6.option.rsa.key _ hash | Key hash |
| icmpv6.option.type | Type |
| icmpv6.ra.cur _ hop _ limit | Cur hop limit |
| icmpv6.ra.reachable _ time | Reachable time |
| icmpv6.ra.retrans _ timer | Retrans timer |
| icmpv6.ra.router _ lifetime | Router lifetime |
| icmpv6.recursive _ dns _ serv | Recursive DNS Servers |
| icmpv6.type | Type |
| http.connection | Connection |
| http.content _ encoding | Content-Encoding |
| http.content _ length | Content length |
| http.content _ type | Content-Type |

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| | |
|---|---|
| http.cookie | Cookie |
| http.date | Date |
| http.file_data | File Data |
| http.host | Host |
| http.last_modified | Last-Modified |
| http.location | Location |
| http.notification | Notification |
| http.proxy_authenticate | Proxy-Authenticate |
| http.request.method | Request Method |
| http.request.uri | Request URI |
| http.request.version | Request Version |
| http.response | Filter all HTTP get responses |
| http.response.code | HTTP |
| http.request or http.response | Filter all HTTP get requests and responses |
| http.server | Server |
| http.set_cookie | Set-Cookie |
| http.transfer_encoding | Transfer-Encoding |
| http.user_agent | User-Agent |
| http.www_authenticate | WWW-Authenticate |
| http.x_forwarded_for | X-Forwarded-For |
| http.user_agent contains Firefox | Find user agents |
| !http.user_agent contains \|\| !http.user_agent contains Chrome | Find user agents |
| frame contains "GET" && !tcp.port==80 | HTTP Get not on port 80 |
| http.response.code>299 && http.response.code<40 | HTTP Redirections |
| http.request.uri matches "\.(exe\|zip\|jar)$" | HTTP .exe,.zip,.jar objects |
| http.request.method in {PUT POST} | HTTP PUT and POST messages |

### 18. BGP Filters

| Wireshark Filters | Description |
|---|---|
| bgp.aggregator_as | Aggregator AS |
| bgp.aggregator_origin | Aggregator origin |
| bgp.as_path | AS Path |
| bgp.cluster_identifier | Cluster identifier |
| bgp.cluster_list | Cluster List |
| bgp.community_as | Community AS |
| bgp.community_value | Community value |
| bgp.local_pref | Local preference |
| bgp.mp_nlri_tnl_id | MP Reach NLRI Tunnel Identifier |
| bgp.mp_reach_nlri_ipv4_prefix | MP Reach NLRI IPv4 prefix |
| bgp.mp_unreach_nlri_ipv4_prefix | MP Unreach NLRI IPv4 prefix |
| bgp.multi_exit_disc | Multiple exit discriminator |
| bgp.next_hop | Next hop |
| bgp.nlri_prefix | NLRI prefix |
| bgp.origin | Origin |
| bgp.originator_id | Originator identifier |
| bgp.type | Type |
| bgp.withdrawn_prefix | Withdrawn prefix |
| bgp.wildcard_route_target | Wildcard route target |
| bgp.vplsbgp.labelblock.size | Label Block Size |
| bgp.vplsbgp.labelblock.offset | Label Block Offset |
| bgp.vplsbgp.labelblock.base | Label Block Base |
| bgp.vplsad.length | Length |
| bgp.update.withdrawn_routes.length | Withdrawn Routes Length |
| bgp.update.withdrawn_routes | Withdrawn Routes |
| bgp.update.path_attribute.multi_exit_disc | Multiple exit discriminator |
| bgp.update.path_attribute.next_hop | Next hop |
| bgp.unknown_safi | Unknown SAFI |
| bgp.unknown_label | Unknown Label VPN |
| bgp.unknown_afi | Unknown Address Family |
| bgp.ssa_value | Value |
| bgp.ssa_type | SSA Type |
| bgp.ssa_t | Transitive bit |
| bgp.ssa_l2tpv3_Unused | Unused |
| bgp.ssa_l2tpv3_session_id | Session ID |
| bgp.ssa_l2tpv3_s | Sequencing bit |
| bgp.ssa_l2tpv3_pref | Preference |
| bgp.ssa_l2tpv3_cookie_len | Cookie Length |
| bgp.route_refresh.subtype | Subtype |
| bgp.route_refresh.orf.entry.ip | ORFEntry IP address |
| bgp.rd | Route Distinguisher |
| bgp.prefix_sid.type_err | BGP Prefix-SID unknown TLV type |
| bgp.prefix_length | Prefix Length |
| bgp.path_attribute.cluster_id | Cluster ID |
| bgp.path_attribute.cluster_list | Cluster List |
| bgp.open.version | Version |
| bgp.open.opt.param.type | Parameter Type |
| bgp.open.holdtime | Hold Time |
| bgp.notify.minor_error_update | Minor error Code (Update Message) |
| bgp.notify.minor_error_unknown | Minor error Code (Unknown) |
| bgp.notify.minor_error_state | Minor error Code (State Machine) |
| bgp.notify.minor_error_open | Minor error Code (Open Message) |
| bgp.nlri_path_id | NLRI path id |
| bgp.mp_unreach_nlri_ipv6_prefix | MP Unreach NLRI IPv6 prefix |
| bgp.mdt_safi_rd | Route Distinguisher |
| bgp.mdt_safi_ipv4_addr | IPv4 Address |
| bgp.mdt_safi_group_addr | Group Address |
| bgp.mcast_vpn_nlri_source_length | Multicast Source Length |
| bgp.marker_invalid | Marker is not all ones |
| bgp.marker | Marker |
| bgp.ls.warn | Link State warning |
| bgp.ls.tlv.unreserved_bandwidth | Unreserved bandwidth TLV |
| bgp.ls.node_flag_bits.abr | ABR Bit |
| bgp.ls.node_flag_bits.external | External Bit |
| bgp.ls.bandwidth_value | Bandwidth |



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| | |
|---|---|
| bgp.ls.error | Link State error |
| bgp.large _ communities | Large Communities |
| bgp.label _ stack | Label Stack |
| bgp.flowspec _ route _ distinguisher | Route Distinguisher |
| bgp.flags.transitive | Transitive |
| bgp.ext _ communities | Carried extended communities |
| bgp.evpn.nlri.rt | Route Type |
| bgp.cap.orf.sendreceive | Send Receive |
| bgp.attr.pmsi.tunnel _ type | Unknown Tunnel type |
| bgp.afi _ type _ not _ supported | AFI Type not supported |
| bgp.ad.pe _ addr | PE Addr |

### 19. TLS Filters

| Wireshark Filters | Description |
|---|---|
| tls | All TLS Packets |
| tls.record.content _ type == 22 | TLS Handshake Packets |
| tls.handshake | Handshake Protocol |
| tls.handshake.cert _ type | Certificate Type |
| tls.handshake.cert _ type.type | Certificate Type |
| tls.handshake.cert _ type.types | Certificate Type List |
| tls.handshake.cert _ type.types _ len | Certificate Type List Length |
| tls.handshake.type == 1 | TLS Client Hello Packets |
| tls.handshake.type == 2 | TLS Server Hello Packets |
| tls.record.content _ type == 21 | TLS Encrypted Alert |
| tls.handshake.cert _ types | Certificate types |
| tls.handshake.cert _ types _ count | Certificate types count |
| tls.handshake.cert _ url.padding | Padding |
| tls.handshake.cert _ url.sha1 | SHA1 Hash |
| tls.handshake.cert _ url.url | URL |
| tls.handshake.cert _ url.url _ hash | URL and Hash |
| tls.handshake.cert _ url.url _ len | URL Length |
| tls.handshake.cert _ url.url _ hash _ len | URL and Hash list Length |
| tls.handshake.cert _ url _ type | Certificate Chain Type |
| tls.handshake.certificate | Certificate |
| tls.handshake.certificate _ length | Certificate Length |
| tls.handshake.certificate _ request _ context | Certificate Request Context |
| tls.handshake.certificate _ request _ context _ length | Certificate Request Context Length |
| tls.handshake.certificates | Certificates |
| tls.handshake.certificates _ length | Certificates Length |
| tls.handshake.challenge | Challenge |
| tls.handshake.challenge _ length | Challenge Length |
| tls.handshake.cipher _ spec _ len | Cipher Spec Length |
| tls.handshake.cipher _ suites _ length | Cipher Suites Length |
| tls.handshake.cipherspec | Cipher Spec |
| tls.handshake.ciphersuite | Cipher Suite |
| tls.handshake.ciphersuites | Cipher Suites |
| tls.handshake.clear _ key _ data | Clear Key Data |
| tls.handshake.clear _ key _ length | Clear Key Data Length |
| tls.handshake.client _ cert _ vrfy. sig | Signature |
| tls.handshake.client _ cert _ vrfy. sig _ len | Signature length |
| tls.handshake.client _ point | Pubkey |
| tls.handshake.client _ point _ len | Pubkey Length |
| tls.handshake.comp _ method | Compression Method |
| Tls.handshake.extensions _ server _ name contains "hack" | TLS contains "hack" in server name |
| tls.alert _ message | Alert Message |
| tls.alert _ message.desc | Description |
| tls.alert _ message.level | Level |
| tls.app _ data | Encrypted Application Data |
| tls.change _ cipher _ spec | Change Cipher Spec Message |
| tls.compress _ certificate. algorithm | Algorithm |
| tls.compress _ certificate. algorithms _ length | Algorithms Length |
| tls.compress _ certificate. compressed _ certificate _ message | Compressed Certificate Message |
| tls.compress _ certificate. compressed _ certificate _ message. length | Length |
| tls.compress _ certificate. uncompressed _ length | Uncompressed Length |
| tls.early _ data.max _ early _ data _ size | Maximum Early Data Size |
| tls.esni.encrypted _ sni | Encrypted SNI |
| tls.esni.encrypted _ sni _ length | Encrypted SNI Length |
| tls.esni.nonce | Nonce |
| tls.esni.record _ digest | Record Digest |
| tls.esni.record _ digest _ length | Record Digest Length |
| tls.esni.suite | Cipher Suite |
| tls.extension.oid _ filters.oid | Certificate Extension OID |
| tls.extension.oid _ filters.oid _ length | Certificate Extension OID Length |
| tls.extension.oid _ filters. values _ length | Certificate Extension Values Length |
| tls.extension.oid _ filters _ length | OID Filters Length |
| tls.extension.psk _ ke _ mode | PSK Key Exchange Mode |
| tls.extension.psk _ ke _ modes _ length | PSK Key Exchange Modes Length |
| tls.handshake.comp _ methods | Compression Methods |
| tls.handshake.comp _ methods _ length | Compression Methods Length |
| tls.handshake.connection _ id | Connection ID |
| tls.handshake.connection _ id _ length | Connection ID Length |
| tls.handshake.dname | Distinguished Name |
| tls.handshake.dname _ len | Distinguished Name Length |
| tls.handshake.dnames | Distinguished Names |
| tls.handshake.dnames _ len | Distinguished Names Length |
| tls.handshake.encrypted _ key | Encrypted Key |
| tls.handshake.encrypted _ key _ length | Encrypted Key Data Length |
| tls.handshake.epms | Encrypted PreMaster |
| tls.handshake.epms _ len | Encrypted PreMaster length |
| tls.handshake.exponent | Exponent |
| tls.handshake.exponent _ len | Exponent Length |
| tls.handshake.extension.data | Data |
| tls.handshake.extension. heartbeat.mode | Mode |

# Ethical Hacking and Countermeasures
## Wireshark Filters Cheat Sheet

EC-Council
Building A Culture Of Security

**Wireshark**
Source: https://www.wireshark.org

Wireshark is a protocol analyzer tool that can capture and interactively browse the traffic running on a computer network. This tool uses Winpcap to capture packets on its supported networks. It captures live network traffic from Ethernet, IEEE 802.11, PPP/HDLC, ATM, Bluetooth, USB, Token Ring, Frame Relay, FDDI networks. The captured files can be programmatically edited via the command-line. A set of filters for customized data displays can be refined using a display filter.

| | |
|---|---|
| tls.handshake.extension.len | Length |
| tls.handshake.extension.type | Type |
| tls.handshake.extensions.cookie | Cookie |
| tls.handshake.extensions.cookie_len | Cookie length |
| tls.handshake.extensions.psk.binders | PSK Binders |
| tls.handshake.extensions.psk.binders_len | PSK Binders length |
| tls.handshake.extensions.psk.identities.length | Identities Length |
| tls.handshake.extensions.psk.identity.identity | Identity |
| tls.handshake.extensions.psk.identity.identity_length | Identity Length |
| tls.handshake.extensions.psk.identity.obfuscated_ticket_age | Obfuscated Ticket Age |
| tls.handshake.extensions.psk.identity.selected | Selected Identity |
| tls.handshake.extensions.supported_version | Supported Version |
| tls.handshake.extensions.supported_versions_len | Supported Versions length |
| tls.handshake.extensions_alpn_len | ALPN Extension Length |
| tls.handshake.extensions_alpn_list | ALPN Protocol |
| tls.handshake.extensions_alpn_str | ALPN Next Protocol |
| tls.handshake.extensions_alpn_str_len | ALPN string length |
| tls.handshake.extensions_ec_point_format | EC point format |
| tls.handshake.extensions_ec_point_formats | EC point formats |
| tls.handshake.extensions_ec_point_formats_length | EC point formats Length |
| tls.handshake.extensions_key_share_client_length | Client Key Share Length |
| tls.handshake.extensions_key_share_group | Group |
| tls.handshake.extensions_key_share_key_exchange | Key Exchange |
| tls.handshake.extensions_key_share_key_exchange_length | Key Exchange Length |
| tls.handshake.extensions_key_share_selected_group | Selected Group |
| tls.handshake.extensions_length | Extensions Length |
| tls.handshake.extensions_npn | Next Protocol |
| tls.handshake.extensions_npn_str_len | Protocol string length |
| tls.handshake.extensions_padding_data | Padding Data |
| tls.handshake.extensions_reneg_info | Renegotiation info |
| tls.handshake.extensions_reneg_info_len | Renegotiation info extension length |
| tls.handshake.extensions_server_name | Server Name |
| tls.handshake.extensions_server_name_len | Server Name length |
| tls.handshake.extensions_server_name_list_len | Server Name list length |
| tls.handshake.extensions_server_name_type | Server Name Type |
| tls.handshake.extensions_status_request_exts_len | Request Extensions Length |
| tls.handshake.extensions_status_request_len | Certificate Status Length |
| tls.handshake.extensions_status_request_list_len | Certificate Status List Length |
| tls.handshake.extensions_status_request_responder_ids_len | Responder ID list Length |
| tls.handshake.extensions_status_request_type | Certificate Status Type |
| tls.handshake.extensions_supported_group | Supported Group |
| tls.handshake.extensions_supported_groups | Supported Groups List |
| tls.handshake.extensions_supported_groups_length | Supported Groups List Length |
| tls.handshake.fragment | Handshake Fragment |
| tls.handshake.fragment.count | Handshake Fragment count |
| tls.handshake.fragments | Reassembled Handshake Fragments |
| tls.handshake.hint | Hint |
| tls.handshake.hint_len | Hint Length |
| tls.handshake.identity | Identity |
| tls.handshake.identity_len | Identity Length |
| tls.handshake.key_arg | Key Argument |
| tls.handshake.key_arg_length | Key Argument Length |
| tls.handshake.key_update.request_update | Key Update Request |
| tls.handshake.length | Length |
| tls.handshake.max_fragment_length | Maximum Fragment Length |
| tls.handshake.md5_hash | MD5 Hash |
| tls.handshake.modulus | Modulus |
| tls.handshake.modulus_len | Modulus Length |
| tls.handshake.npn_padding | Padding |
| tls.handshake.npn_padding_len | Padding Length |
| tls.handshake.npn_selected_protocol | Selected Protocol |
| tls.handshake.npn_selected_protocol_len | Selected Protocol Length |
| tls.handshake.ocsp_response_len | OCSP Response Length |
| tls.handshake.ocsp_response_list_len | OCSP Response List Length |
| tls.handshake.random | Random |
| tls.handshake.random_bytes | Random Bytes |
| tls.handshake.random_time | GMT Unix Time |
| tls.handshake.reassembled_in | Reassembled Handshake Message in frame |
| tls.handshake.server_curve_type | Curve Type |
| tls.handshake.server_named_curve | Named Curve |
| tls.handshake.server_point | Pubkey |
| tls.handshake.server_point_len | Pubkey Length |
| tls.handshake.session_id | Session ID |
| tls.handshake.session_id_hit | Session ID Hit |
| tls.handshake.session_id_length | Session ID Length |
| tls.handshake.session_id_length.error | Session ID length error |
| tls.handshake.session_ticket | Session Ticket |
| tls.handshake.session_ticket_age_add | Session Ticket Age Add |
| tls.handshake.session_ticket_length | Session Ticket Length |
| tls.handshake.session_ticket_lifetime_hint | Session Ticket Lifetime Hint |



CEH: 1 year access to 12 Capture-The-Flag (CTF) challenges

**Wireshark**
Source: https://www.wireshark.org

Wireshark is a protocol analyzer tool that can capture and interactively browse the traffic running on a computer network. This tool uses Winpcap to capture packets on its supported networks. It captures live network traffic from Ethernet, IEEE 802.11, PPP/HDLC, ATM, Bluetooth, USB, Token Ring, Frame Relay, FDDI networks. The captured files can be programmatically edited via the command-line. A set of filters for customized data displays can be refined using a display filter.

| | |
|---|---|
| tls.handshake.session_ticket_nonce | Session Ticket Nonce |
| tls.handshake.session_ticket_nonce_length | Session Ticket Nonce Length |
| tls.handshake.status_request.undecoded | Responder ID list or Request Extensions are not implemented, contact Wireshark developers if you want this to be supported |
| tls.handshake.verify_data | Verify Data |
| tls.heartbeat_message | Heartbeat Message |
| tls.heartbeat_message.padding | Payload Length |
| tls.heartbeat_message.payload | Payload Length |
| tls.ignored_unknown_record | Ignored Unknown Record |
| tls.malformed.buffer_too_small | Malformed message, not enough data is available |
| tls.malformed.trailing_data | Undecoded trailing data is present |
| tls.malformed.vector_length | The variable vector length is outside the permitted range |
| tls.quic.initial_version | Initial Version |
| tls.quic.len | Parameters Length |
| tls.quic.negotiated_version | Negotiated Version |
| tls.quic.parameter | Parameter |
| tls.quic.parameter.ack_delay_exponent | ack_delay_exponent |
| tls.quic.parameter.active_connection_id_limit | Active Connection ID Limit |
| tls.quic.parameter.idle_timeout | idle_timeout |
| tls.quic.parameter.initial_max_data | initial_max_data |
| tls.reassembled.data | Reassembled PDU data |
| tls.reassembled.length | Reassembled PDU length |
| tls.reassembled_in | Reassembled PDU in frame |
| tls.record | Record Layer |
| tls.record.content_type | Content Type |
| tls.record.is_escape | Is Escape |
| tls.record.length | Length |
| tls.record.length.invalid | Record fragment length is too small or too large |
| tls.record.opaque_type | Opaque Type |
| tls.record.padding_length | Padding Length |
| tls.record.version | Version |
| tls.record_size_limit | Record Size Limit |
| tls.resumed | This session reuses previously negotiated keys (Session resumption) |
| tls.sct.sct_extensions | Extensions |
| tls.sct.sct_extensions_length | Extensions length |
| tls.sct.sct_length | Serialized SCT Length |
| tls.sct.sct_logid | Log ID |
| tls.sct.sct_signature | Signature |
| tls.sct.sct_signature_length | Signature Length |
| tls.sct.sct_timestamp | Timestamp |
| tls.sct.sct_version | SCT Version |
| tls.sct.scts_length | Serialized SCT List Length |
| tls.segment | TLS segment |
| tls.segment.count | Segment count |
| tls.segment.data | TLS segment data |
| tls.segment.error | Reassembling error |
| tls.segment.multipletails | Multiple tail segments found |
| tls.segment.overlap | Segment overlap |
| tls.segment.overlap.conflict | Conflicting data in segment overlap |
| tls.segment.toolongfragment | Segment too long |
| tls.segments | Reassembled TLS segments |
| tls.unexpected_message | Unexpected message |

### 20. RIPv2 Filters

| Wireshark Filters | Description |
|---|---|
| rip.auth.passwd | Password |
| rip.auth.type | Authentication type |
| rip.auth_data_len | Auth Data Len |
| rip.authentication_data | Authentication Data |
| rip.command | Command |
| rip.digest_offset | Digest Offset |
| rip.family | Address Family |
| rip.ip | IP Address |
| rip.key_id | Key ID |
| rip.metric | Metric |
| rip.netmask | Netmask |
| rip.next_hop | Next Hop |
| rip.route_tag | Route Tag |
| rip.routing_domain | Routing Domain |
| rip.seq_num | Seq Number |
| rip.unknown_address_family | Unknown address family |
| rip.zero_padding | Zero adding |
| rip.version | Version |

### 21. OSPF and OSPFv2 Filters

| Wireshark Filters | Description |
|---|---|
| ospf.advrouter | Advertising Router |
| ospf.area_id | Area ID |
| ospf.at | OSPF Authentication Trailer |
| ospf.at.auth_data | Authentication Data |
| ospf.at.auth_data_len | Authentication Data Length |
| ospf.at.auth_type | Authentication Type |
| ospf.at.cryto_seq_nbr | Cryptographic Sequence Number |
| ospf.at.reserved | Reserved |
| ospf.at.sa_id | Security Association Identifier (SA ID) |
| ospf.auth.crypt.data | Auth Crypt Data |
| ospf.auth.crypt.data_length | Auth Crypt Data Length |
| ospf.auth.crypt.key_id | Auth Crypt Key id |
| ospf.auth.crypt.seq_nbr | Auth Crypt Sequence Number |
| ospf.auth.none | Auth Data (none) |
| ospf.auth.simple | Auth Data (Simple) |
| ospf.auth.type | Auth Type |
| ospf.auth.unknown | Auth Unknown |
| ospf.checksum | Checksum |
| ospf.db.dd_sequence | DD Sequence |
| ospf.db.interface_mtu | Interface MTU |
| ospf.dbd | DB Description |
| ospf.dbd.i | (I) Init |
| ospf.dbd.m | (M) More |
| ospf.dbd.ms | (MS) Master |



92% Reported That the C|EH Boosted Their Confidence

# Ethical Hacking and Countermeasures
## Wireshark Filters Cheat Sheet

EC-Council
Building A Culture Of Security

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Source: https://www.wireshark.org

Wireshark is a protocol analyzer tool that can capture and interactively browse the traffic running on a computer network. This tool uses Winpcap to capture packets on its supported networks. It captures live network traffic from Ethernet, IEEE 802.11, PPP/HDLC, ATM, Bluetooth, USB, Token Ring, Frame Relay, FDDI networks. The captured files can be programmatically edited via the command-line. A set of filters for customized data displays can be refined using a display filter.

| | |
|---|---|
| ospf.dbd.r | (R) OOBResync |
| ospf.dynhostname | Dynamic Hostname |
| ospf.header | OSPF Header |
| ospf.hello | OSPF Hello Packet |
| ospf.hello.active _ neighbor | Active Neighbor |
| ospf.hello.backup _ designated _ router | Backup Designated Router |
| ospf.hello.designated _ router | Designated Router |
| ospf.hello.hello _ interval | Hello Interval [sec] |
| ospf.hello.interface _ id | Interface ID |
| ospf.hello.network _ mask | Network Mask |
| ospf.hello.router _ dead _ interval | Router Dead Interval [sec] |
| ospf.hello.router _ priority | Router Priority |
| ospf.instance _ id | Instance ID |
| ospf.link _ state _ id | Link State ID |
| ospf.lls.checksum | Checksum |
| ospf.lls.data _ length | LLS Data Length |
| ospf.lls.ext.options | Options |
| ospf.lls.ext.options.lr | (LR) LSDB Resynchronization |
| ospf.lls.ext.options.rs | (RS) Restart Signal |
| ospf.ls.metric | Metric |
| ospf.ls.number _ of _ lsas | Number of LSAs |
| ospf.lsa | LS Type |
| ospf.lsa.age | LS Age (seconds) |
| ospf.mpls.routerid | MPLS/TE Router ID |
| ospf.mpls.shared _ risk _ link _ group | Shared Risk Link Group |
| ospf.mpls.sonet.sdh | SONET/SDH |
| ospf.mpls.starting | Starting n |
| ospf.mpls.switching _ type | Switching Type |
| ospf.mpls.te _ metric | MPLS/TE Metric |
| ospf.mpls.type | Type |
| ospf.msg | Message Type |
| ospf.msg.dbdesc | Database Description |
| ospf.msg.hello | Hello |
| ospf.msg.lsack | Link State Adv Acknowledgement |
| ospf.msg.lsreq | Link State Adv Request |
| ospf.msg.lsupdate | Link State Adv Update |
| ospf.oif.encoding | Encoding |
| ospf.oif.local _ node _ id | Local Node ID |
| ospf.oif.node _ id | Node ID |
| ospf.oif.remote _ node _ id | Remote Node ID |
| ospf.oif.signal _ type | Signal Type |
| ospf.oif.switching _ cap | Switching Cap |
| ospf.oif.tna _ addr | TNA Addr |
| ospf.oif.tna _ addr.ipv4 | TNA Addr |
| ospf.oif.tna _ addr.ipv6 | TNA Addr |
| ospf.oif.tna _ addr _ length | Addr Length |
| ospf.packet _ length | Packet Length |
| ospf.pad _ bytes | Pad Bytes |
| ospf.prefix _ length | PrefixLength |
| ospf.reserved | Reserved |
| ospf.reserved.not _ zero | incorrect, should be 0 |
| ospf.ri.mbz | MBZ |
| ospf.ri.options | RI Options |
| ospf.ri.options.ete | (ETE) Experimental TE |
| ospf.ri.options.grc | (GRC) Graceful Restart |
| ospf.ri.options.grh | (GRH) Graceful Restart Helper |
| ospf.ri.options.p2plan | (P2PLAN) Point-to-point over LAN |
| ospf.ri.options.srs | Stub Router Support |
| ospf.ri.options.tes | (TES) Traffic Engineering |
| ospf.tlv.adjsid.flags | Flags |
| ospf.tlv.adjsid.flags.b | (B) Backup Flag |
| ospf.tlv.adjsid.flags.g | (G) Group Flag |
| ospf.tlv.adjsid.flags.p | (P) Persistent Flag |
| ospf.tlv.adjsid.flags.unknown | (*) Unknown Flag |
| ospf.tlv.adjsid.flags.v | (V) Value/Index Flag |
| ospf.tlv.extlink.mt _ id | Multi-Topology ID |
| ospf.tlv.extlink.nbr | Neighbor ID |
| ospf.tlv.extlink.subtlv _ type | TLV Type |
| ospf.tlv.extlink.tlv _ type | TLV Type |
| ospf.tlv.extlink.weight | Weight |
| ospf.tlv.extpfx.af | Address Family |
| ospf.tlv.extpfx.flags | Flags |
| ospf.tlv.extpfx.flags.a | (A) Attach Flag |
| ospf.tlv.extpfx.flags.n | (N) Node Flag |
| ospf.tlv.extpfx.flags.unknown | (*) Unknown Flag |
| ospf.tlv.extpfx.rotuetype | Route Type |
| ospf.tlv.extpfx.subtlv _ type | TLV Type |
| ospf.tlv.extpfx.tlv _ type | TLV Type |
| ospf.tlv.extpfx _ range.flags | Flags |
| ospf.tlv.extpfx _ range.flags.ia | (IA) Inter-Area Flag |
| ospf.tlv.extpfx _ range.flags. unknown | (*) Unknown Flag |
| ospf.tlv.igp _ msd _ type | MSD Type |
| ospf.tlv.igp _ msd _ value | MSD Value |
| ospf.tlv.local _ interface _ id | Local Interface ID |
| ospf.tlv.pfxsid.flags | Flags |
| ospf.tlv.pfxsid.flags.e | (E) Explicit-Null Flag |
| ospf.tlv.pfxsid.flags.l | (L) Local/Global Flag |
| ospf.tlv.pfxsid.flags.m | (M) Mapping Server Flag |
| ospf.tlv.pfxsid.flags.np | (NP) No-PHP Flag |
| ospf.tlv.pfxsid.flags.unknown | (*) Unknown Flag |
| ospf.tlv.pfxsid.flags.v | (V) Value/Index Flag |
| ospf.tlv.preference | Preference |
| ospf.tlv.range _ size | Range Size |
| ospf.tlv.remote _ interface _ id | Remote Interface ID |
| ospf.tlv.remote _ ipv4 _ address | Remote IPv4 Address |
| ospf.tlv.sid _ label | SID/Label |
| ospf.tlv.sidlabel _ range.type | TLV Type |
| ospf.tlv.unknown | Unknown TLV |
| ospf.tlv _ length | TLV Length |
| ospf.tlv _ type | TLV Value |
| ospf.tlv _ type.opaque | TLV Value |
| ospf.tlv _ value | TLV Value |



97% Stated That the Skills They Acquired in C|EH Helped Safeguard Their Organizations

# Ethical Hacking and Countermeasures
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| | |
|---|---|
| ospf.unknown _ link _ subtype | Unknown Link sub-TLV |
| ospf.unknown _ text | Text in the Unknown TLV |
| ospf.srcrouter | Source OSPF Router |
| ospf.lsa.asbr | Summary LSA (ASBR) |
| ospf.lsa.asbr.netmask | Netmask |
| ospf.lsa.asext | AS-External LSA (ASBR) |
| ospf.lsa.asext.extrttag | External Route Tag |
| ospf.lsa.asext.fwdaddr | Forwarding Address |
| ospf.lsa.asext.netmask | Netmask |
| ospf.lsa.asext.type | External Type |
| ospf.lsa.attr | External Attributes LSA |
| ospf.lsa.bc _ error | BC error |
| ospf.lsa.chksum | Checksum |
| ospf.lsa.donotage | Do Not Age Flag |
| ospf.lsa.id | Link State ID |
| ospf.lsa.invalid _ length | Invalid length |
| ospf.lsa.length | Length |
| ospf.lsa.member | Group Membership LSA |
| ospf.lsa.mpls | MPLS Traffic Engineering LSA |
| ospf.lsa.network | Network LSA |
| ospf.lsa.network.attchrtr | Attached Router |
| ospf.lsa.network.netmask | Netmask |
| ospf.lsa.nssa | NSSA AS-External LSA |
| ospf.lsa.number _ of _ links | Number of Links |
| ospf.lsa.opaque | Opaque LSA |
| ospf.lsa.router | Router LSA |
| ospf.lsa.router.linkdata | Link Data |
| ospf.lsa.router.linkid | Link ID |
| ospf.lsa.router.linktype | Link Type |
| ospf.lsa.router.metric0 | 0 Metric |
| ospf.lsa.router.nummetrics | Number of Metrics |
| ospf.lsa.seqnum | Sequence Number |
| ospf.lsa.summary | Summary LSA (IP Network) |
| ospf.lsa.tos | TOS |
| ospf.lsa.tos _ missing | Blocks missing |
| ospf.lsa.unknown _ type | Unknown LSA Type |
| ospf.lsa _ sa | SR-Algorithm |
| ospf.lsid.opaque _ id | Link State ID Opaque ID |
| ospf.lsid _ opaque _ type | Link State ID Opaque Type |
| ospf.lsid _ te _ lsa.instance | Link State ID TE-LSA Instance |
| ospf.lsid _ te _ lsa.reserved | Link State ID TE-LSA Reserved |
| ospf.metric | Metric |
| ospf.mpls.action | Action |
| ospf.mpls.bandwidth.type | Bandwidth Type |
| ospf.mpls.bc | BC |
| ospf.mpls.bc.model _ id | MPLS/DSTE Bandwidth Constraints Model Id |
| ospf.mpls.bitmap | Bitmap |
| ospf.mpls.cs | Channel Spacing |
| ospf.mpls.effective | No. of effective. Bits |
| ospf.mpls.encoding | Encoding |
| ospf.mpls.grid | Grid |
| ospf.mpls.group | MPLS/TE Group |
| ospf.mpls.interface _ mtu | Interface MTU |
| ospf.mpls.length | Length |
| ospf.mpls.link _ max _ bw | Link Max BW |
| ospf.mpls.linkcolor | MPLS/TE Link Resource Class/Color |
| ospf.mpls.linkid | MPLS/TE Link ID |
| ospf.mpls.linktype | MPLS/TE Link Type |
| ospf.mpls.local _ addr | MPLS/TE Local Interface Address |
| ospf.mpls.local _ id | MPLS/TE Local Interface Index |
| ospf.mpls.minimum _ lsp _ bandwidth | Minimum LSP bandwidth |
| ospf.mpls.n | Starting n |
| ospf.mpls.num.labels | Num Labels |
| ospf.mpls.pri | Pri (or TE-Class) |
| ospf.mpls.priority | Priority |
| ospf.mpls.protection _ capability | Protection Capability |
| ospf.mpls.remote _ addr | MPLS/TE Remote Interface Address |
| ospf.mpls.remote _ id | MPLS/TE Remote Interface Index |
| ospf.v2.grace | Grace TLV |
| ospf.v2.grace.ip | Restart IP |
| ospf.v2.grace.period | Grace Period |
| ospf.v2.grace.reason | Restart Reason |
| ospf.v2.lls.auth _ data | Auth Data |
| ospf.v2.lls.sequence _ number | Sequence number |
| ospf.v2.options | Options |
| ospf.v2.options.dc | (DC) Demand Circuits |
| ospf.v2.options.dn | (E) External Routing |
| ospf.v2.options.l | (L) LLS Data block |
| ospf.v2.options.mc | (MC) Multicast |
| ospf.v2.options.mt | (MT) Multi-Topology Routing |
| ospf.v2.options.n | (N) NSSA |
| ospf.v2.options.p | (P) Propagate |
| ospf.v2.router.lsa.flags | Flags |
| ospf.v2.router.lsa.flags.b | (B) Area border router |
| ospf.v2.router.lsa.flags.e | (E) AS boundary router |
| ospf.v2.router.lsa.flags.n | (N) flag |
| ospf.v2.router.lsa.flags.v | (V) Virtual link endpoint |
| ospf.v2.router.lsa.flags.w | (W) Wild-card multicast receiver |

### 22. OSPFv3/IP v6 Filters

| Wireshark Filters | Description |
|---|---|
| ospf.v3.as.external.flags | Flags |
| ospf.v3.as.external.flags.e | (E) External Metric |
| ospf.v3.as.external.flags.f | (F) Forwarding Address |
| ospf.v3.as.external.flags.t | (T) External Route Tag |
| ospf.v3.address _ prefix.ipv4 | Address Prefix |
| ospf.v3.address _ prefix.ipv6 | Address Prefix |
| ospf.v3.lls.dropped _ neighbor | Dropped Neighbor |
| ospf.v3.lls.drop.tlv | Neighbor Drop TLV |
| ospf.v3.lls.ext.options | Options |
| ospf.v3.lls.ext.options.lr | (LR) LSDB Resynchronization |



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# Ethical Hacking and Countermeasures
## Wireshark Filters Cheat Sheet

EC-Council
Building A Culture Of Security

**Wireshark**
Source: https://www.wireshark.org

Wireshark is a protocol analyzer tool that can capture and interactively browse the traffic running on a computer network. This tool uses Winpcap to capture packets on its supported networks. It captures live network traffic from Ethernet, IEEE 802.11, PPP/HDLC, ATM, Bluetooth, USB, Token Ring, Frame Relay, FDDI networks. The captured files can be programmatically edited via the command-line. A set of filters for customized data displays can be refined using a display filter.

| | |
|---|---|
| ospf.v3.lls.ext.options.rs | (RS) Restart Signal |
| ospf.v3.lls.willingness.tlv | Extended Options TLV, Willingness TLV |
| ospf.v3.lls.extended _ options | Extended Options |
| ospf.v3.options | Options |
| ospf.v3.lls.ext.options.tlv | Extended Options TLV |
| ospf.v3.lls.fsf.tlv | Full State For TLV |
| ospf.v3.lls.full _ state _ for | Full State For |
| ospf.v3.lls.ll _ id | Local Interface ID |
| ospf.v3.lls.neighbor | Neighbor |
| ospf.v3.lls.relay.added | Relays Added |
| ospf.v3.lls.relay.options | Options |
| ospf.v3.lls.relay.options.a | (A) Always |
| ospf.v3.lls.relay.options.n | (N) Never |
| ospf.v3.lls.relay.tlv | Active Overlapping Relays TLV |
| ospf.v3.lls.request _ from | Request From |
| ospf.v3.lls.rf.tlv | Request From TLV |
| ospf.v3.lls.state.options | Options |
| ospf.v3.lls.state.options.a | (A) Answer |
| ospf.v3.lls.state.options.n | (N) Incomplete |
| ospf.v3.lls.state.options.r | (R) Resuest |
| ospf.v3.lls.state.scs | SCS Number |
| ospf.v3.lls.state.tlv | State Check Sequence TLV |
| ospf.v3.lls.willingness | Willingness |
| ospf.v3.lsa | LS Type |
| ospf.v3.lsa.asext | AS-External-LSA |
| ospf.v3.lsa.attached _ router | Attached Router |
| ospf.v3.lsa.destination _ router _ id | Destination Router ID |
| ospf.v3.lsa.do _ not _ age | Do Not Age |
| ospf.v3.lsa.external _ route _ tag | External Route Tag |
| ospf.v3.lsa.fc | Function Code |
| ospf.v3.lsa.forwarding _ address. ipv4 | Forwarding Address |
| ospf.v3.lsa.forwarding _ address. ipv6 | Forwarding Address |
| ospf.v3.lsa.interface _ id | Interface ID |
| ospf.v3.lsa.interprefix | Inter-Area-Prefix-LSA |
| ospf.v3.lsa.interrouter | Inter-Area-Router-LSA |
| ospf.v3.lsa.intraprefix | Intra-Area-Prefix-LSA |
| ospf.v3.lsa.link | Link-LSA |
| ospf.v3.lsa.link _ local _ interface _ address.ipv4 | Link-local Interface Address |
| ospf.v3.lsa.link _ local _ interface _ address.ipv6 | Link-local Interface Address |
| ospf.v3.lsa.member | Group-Membership-LSA |
| ospf.v3.lsa.neighbor _ interface _ id | Neighbor Interface ID |
| ospf.v3.lsa.neighbor _ router _ id | Neighbor Router ID |
| ospf.v3.lsa.network | Network-LSA |
| ospf.v3.lsa.nssa | NSSA-LSA |
| ospf.v3.lsa.num _ prefixes | # prefixes |
| ospf.v3.lsa.opaque | Router Information Opaque-LSA |
| ospf.v3.lsa.referenced _ advertising _ router | Referenced Advertising Router |
| ospf.v3.lsa.referenced _ link _ state _ id | Referenced Link State ID |
| ospf.v3.lsa.referenced _ ls _ type | Referenced LS type |
| ospf.v3.lsa.router | Router-LSA |
| ospf.v3.lsa.router _ priority | Router Priority |
| ospf.v3.lsa.s12 | Flooding Scope |
| ospf.v3.lsa.type | Type |
| ospf.v3.lsa.u | LSA Handling |
| ospf.v3.prefix.options | PrefixOptions |
| ospf.v3.prefix.options.la | (LA) Local Address |
| ospf.v3.prefix.options.mc | (MC) Multicast |
| ospf.v3.prefix.options.nu | (NU) NoUnicast |
| ospf.v3.prefix.options.p | (P) Propagate |
| ospf.v3.router.lsa.flags | Flags |
| ospf.v3.router.lsa.flags.b | (B) Area border router |
| ospf.v3.router.lsa.flags.e | (E) AS boundary router |
| ospf.v3.router.lsa.flags.v | (V) Virtual link endpoint |
| ospf.v3.router.lsa.flags.w | (W) Wild-card multicast receiver |
| ospf.version | Version |

### 23.Common Filtering Commands

| Wireshark Filters | Description |
|---|---|
| ip.addr == 10.10.50.1/24 | Filter subnet: display traffic from the subnet, be its source or destination |
| tcp.port == 25 | Filter by TCP port |
| tcp.dstport == 23 | Filter by destination TCP port |
| ip.addr == 10.10.50.1 and Tcp.port == 25 | Filter by IP address and port |
| http.host == "host name" | Filter by URL |
| frame.time >= "June 02, 2019 18:04:00" | Filter by timestamp |
| tcp.flags.syn == 1 or (tcp.seq==1 and tcp.ack==1 and tcp.len==0 and tcp.analysis.initial _ rtt) | Filter SYN flag; Filter three-way handshake |
| tcp.flags.syn == 1 and tcp.flags. ack == 0 | Filter SYN flag; Detecting SYN Floods |
| wlan.fc.type _ subtype = 0x08 | Wireshark Beacon Filter |
| eth.dst == ff:ff:ff:ff:ff:ff | Wireshark broadcast filter |
| (eth.dst[0] & 1) | Wireshark Multicast filter |
| ip.host = hostname | Host name filter |
| eth.addr == 00:70:f4:23:18:c4 | MAC address filter |
| tcp.flags.reset == 1 | RST flag filter |
| !ip.addr ==192.168.0.1 | Exclude IP address: remove traffic from and to IP address |
| ip.addr == 192.168.0.1/24 and ip.addr == 192.168.1.1/24 | Display traffic between two specific subnet |
| frame contains "(attachment\|tar\|exe\|zip\|pdf)" | Find files by type |
| tcp contains <Keyword> | |
| frame contains <Keyword> | Find traffic based on keyword |



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# Ethical Hacking and Countermeasures
## Wireshark Filters Cheat Sheet

**Wireshark**
Source: https://www.wireshark.org

Wireshark is a protocol analyzer tool that can capture and interactively browse the traffic running on a computer network. This tool uses Winpcap to capture packets on its supported networks. It captures live network traffic from Ethernet, IEEE 802.11, PPP/HDLC, ATM, Bluetooth, USB, Token Ring, Frame Relay, FDDI networks. The captured files can be programmatically edited via the command-line. A set of filters for customized data displays can be refined using a display filter.

### 24. Filtering Packets

| Operator | Description |
|---|---|
| `and or &&` | Logical AND<br>All the conditions should match |
| `or or \|\|` | Logical OR<br>Either all or one of the conditions should match |
| `xor or ^^` | Logical XOR<br>exclusive alternation – Only one of the two conditions should match not both |
| `not or !` | Logical NOT<br>Not equal to |
| `[n] […]` | Substring operator<br>Filter a specific word or text |

### 25. Logical Operators

| Operators | Description |
|---|---|
| `No.` | Frame number from the beginning of the packet capture |
| `Time` | Seconds from the first frame |
| `Source (src)` | Source address, commonly an IPv4, IPv6 or Ethernet address |
| `Destination (dst)` | Destination address |
| `Protocol` | The protocol used in the Ethernet frame, IP packet, or TCP segment |
| `Length` | Length of the frame in bytes |

### 26. Default Columns in Packet Capture Output

| Columns | Description |
|---|---|
| `No.` | Frame number from the beginning of the packet capture |
| `Time` | Seconds from the first frame |
| `Source (src)` | Source address, commonly an IPv4, IPv6 or Ethernet address |
| `Destination (dst)` | Destination address |
| `Protocol` | The protocol used in the Ethernet frame, IP packet, or TCP segment |
| `Length` | Length of the frame in bytes |

### 27. Display Window Keyboard Shortcuts

| Accelerator | Description |
|---|---|
| `Tab or Shift+Tab` | Move between screen elements, e.g. from the toolbars to the packet list to the packet detail |
| `Alt+→ or Option+→` | Move to the next packet in the selection history |
| ↓ | Move to the next packet or detail item |
| ↑ | Move to the previous packet or detail item |
| → | In the packet detail, opens the selected tree item |
| `Shift+→` | In the packet detail, opens the selected tree item and all its subtrees |
| `Ctrl+ ↓ or F8` | Move to the next packet, even if the packet list isn't focused |
| `Ctrl+ ↑ or F7` | Move to the previous packet, even if the packet list isn't focused |
| `Ctrl+←` | In the packet detail, closes all tree items |
| `Ctrl+→` | In the packet detail, opens all tree items |
| `Ctrl+.` | Move to the next packet of the conversation (TCP, UDP or IP) |
| `Ctrl+,` | Move to the previous packet of the conversation (TCP, UDP or IP) |
| `Backspace` | In the packet detail, jumps to the parent node |
| `Return or Enter` | In the packet detail, toggles the selected tree item |

### 28. Other Commands

| Commands | Description |
|---|---|
| `tftp \|\| irc \|\| bittorrent` | smb2.cmd==3 or smb2.cmd==5 |
| `Hated Apps` | |
| `frame[100-199] contains "nessus"` | Frame offset 100-199 contains "nessus" in lc |
| `frame[100-199] matches "nessus"` | Frame offset 100-199 contains "nessus" in uc/lc |
| `http.user_agent contains "Nmap"` | Suspected nmap traffic (case sensitive) |
| `frame matches "join #"` | IRC Joins |
| `ftp.request.command=="USER" && tcp.len>50` | Long FTP Username |



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